SCHOOL OF PHYSICAL SCIENCES

NATIONAL INSTITUTE OF SCIENCE EDUCATION AND RESEARCH

SEVENTH SEMESTER LAB PROJECT REPORT

DIODE PUMPED Nd-YAG LASER:

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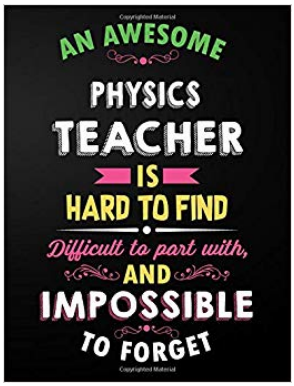

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1 Abstract

The main aim of this project is to understand the principles behind the working of a laser and then using it to create one of the Solid State Laser called the Nd-YAG Laser. This uses Nd-YAG crystal as gain medium and diode laser as source for optical pumping. The laser characteristics of both the laser diode and the Nd-YAG were studied and a threshold current of 144 mA and 140 mA is obtained respectively. This project then explores the relationship between output power and injection current and temperature for both the diode and Nd-YAG lasers. I then go on to calculate the mean life time of the electrons in the meta-stable state, ${}^4F_{\frac{3}{2}}$, which is found to be 263.5 μs The project then further goes on to observe a non-steady state phenomenon (when the laser is switched on or laser in pulsed mode) called spiking wherein output voltages which are hundredfold than the steady state voltages are momentarily achieved. However, we observed a time lag between the first spiking of the voltage and switching on the laser. Appropriate analysis showed that above the threshold injection current, this time was inversely proportional to the cube root of injection current-threshold current. The project further tries to understand the one of the non-linear optical phenomenon called second harmonic generation. Using a KTP crystal 1064 nm light is converted into green light of 532 nm using the second harmonic generation. It is concluded that high electric fields is required for this process to occur since no green light was produced until the crystal was kept in the resonant cavity. Additionally, the relation $P_{2\nu} \propto P_{\nu}^2$ is verified. The project ends by observing the TEM Modes of the second harmonic. Various modes are observed and their intensity profiles are plotted on a 3d-plot. TEM_{00} has been a bit more explored by me. This mode looks like a bright spot on the screen and the intensity profile matches with a 2-dimensional Gaussian distribution with a fairly broad peak implying that the point with maximum intensity is not very sharp when compared to nearby points. This is further backed by the observation that I was not able to clearly tell the centre of the TEM_{00} mode. This intensity profile is expected theoretically since this mode has no radial and angular modes and hence must have a symmetric distribution over the illuminating area.

2 Introduction[7]

LASER stands for Light Amplification By Stimulated Emission Of Radiation and the name arises from how laser's themselves are produced. Before going into the how lasers work, let us see why lasers are useful. There are three unique properties that make the lasers useful.

1. Linewidth- The purity of a laser referred to as linewidth can be very narrow. In Laymann's terms this is the measure of what frequencies are contained in the emitted light. The narrower the linewidth, the closer is the spread of frequency of emitted light from the desired light. Laser in theory are exactly monochromatic. However in practise, there is spread of frequencies form $\nu - d\nu$ to $\nu + d\nu$.

2. Coherence- Light emitted by Laser is coherent, This means that all beams are polarised in the same direction and are in the same phase. This property is required during many optical experiments including interference and diffraction experiments.

3. Power- : Lasers make it possible to deliver high intensity light onto a small area.

After getting to know about the unique properties of Laser, let us understand how a laser actually works. Every Laser has 3 key working processes:

- Stimulated Absorption: We provide an e^- in lower excited state, E_1, generally ground state with a photon of exact energy to jump to a higher excited state, E_2. Remember the states E1 and E2 are bound states of the atom. It is simply the process:

$$e^- \ (E_1) + \gamma \ (h\nu) \rightarrow e^- \ (E_2) \tag{2.0.1}$$

- Spontaneous Emission: Since the higher energy level is unstable, the e^- tries to fall back to the lower energy state. The process is simply:

$$e^- \ (E_2) \rightarrow e^- \ (E_1) + \gamma \ (h\nu) \tag{2.0.2}$$

  The difference in the energy levels determine the frequency of the emitted photon. If frequency is in the visible range we see the light i.e. perceive it as color. The force that pushes the e^- back to the ground state are small perturbations in the vacuum effect (Quantum Mechanical Effects). Since the emission is spontaneous, we call it spontaneous emission. Classically, the situation can be thought of as excited state being a state of unstable equilibrium position and even the smallest forces can make a large change to the system by driving it into the stable equilibrium state.

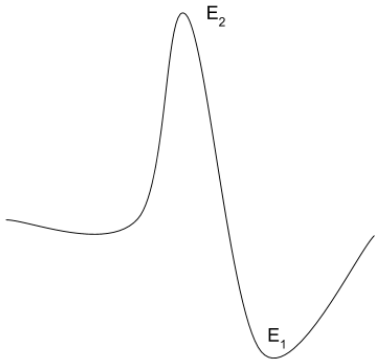


Figure 1: Classically E_2 can be thought of as position of unstable equilibrium and even the smallest perturbations allows the system to go to stable state E_1

- Stimulated Emission: In this process the photon interacts with an e^- that is already in the excited state. The photon field provides the necessary perturbation that drives the electron to go back to the initial stable energy state. This is quite counter-intuitive as one generally expects the e^- to absorb the photon and go to another higher excited state or get ionized. However, this is a valid quantum mechanical process and the process can be shown as:

$$e^- \ (E_2) + \gamma \ (h\nu) \rightarrow e^- \ (E_1) + \gamma \ (h\nu) + \gamma \ (h\nu) \tag{2.0.3}$$

  The photon emitted will be the same (identical) as the one that stimulated it. Both will be coherent with each other i.e. have same phase and polarization. This is in contrast to two e^- 's

undergoing spontaneous emission where it is unlikely that the two emitted photons are travelling in the same direction or have the phase.

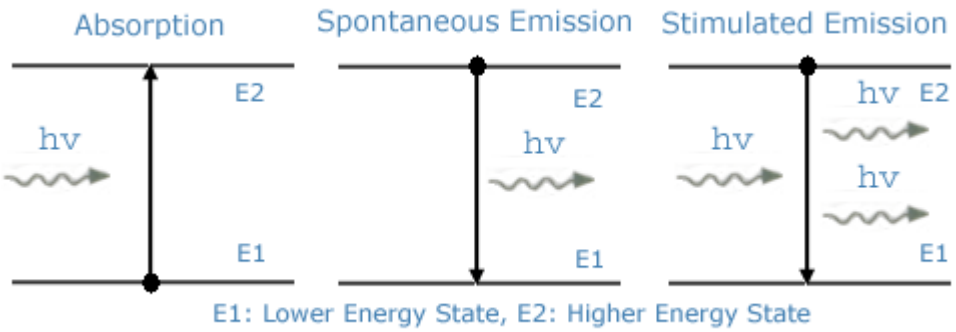


Figure 2: Working principles of a Laser[3]

But for e^- to go to stimulated emission and not spontaneous emission, enough time must be available to the e^- to remain in the excited state. This basically means that the life time of the electron in the excited state should be large enough to allow the stimulated emission. Generally: $\tau_{e^-,ES} \approx 100$ ns which is too short for stimulated emission.
However some materials have states called meta-stable states. These are the excited states with slightly lower energy than the usual excited states. Generally: $\tau_{e^-,MS} \approx 1$ ms which is good enough for passing photon to undergo for stimulated emission. Now if a photon stimulates it (which comes from spontaneous emission from meta-stable state to lower energy state with E'=$h\nu'$), it will produce coherent photon with energy $h\nu'$ which make up the laser beam. Clearly the photons being lased have a smaller frequency than the photons being used to pump the electrons to higher excited state.
We also want lots of e^- in the meta-stable state (more so than the ground state) in order for them to be in a situation where stimulated emission can occur. This is known as creating the population inversion where the number of electrons in the meta-stable state is higher than the lower energy state (generally ground state).
Now, we have created two coherent photons. But we need amplification as we don't need just two puny photons. We want millions and millions of them, all coherent. Since, it is not practical to create extremely long laser setups so the solution is to put the laser gain medium in a cavity. For amplification, we desire constructive interference in our cavity to take place (hence also called resonant cavity). The cavity is generally produced by placing the laser gain medium between two mirrors one of which is full reflecting and the other is partially transmitting to the photons being lased. The waves travelling perpendicular to the plane of mirror will be reflected back and forth to produce interference. This will lead to the production of standing waves in the cavity. The frequencies that will be amplified by the cavity is given by the formula:

$$f = \frac{nv}{2L} \tag{2.0.4}$$

where f is frequency of wave, v is speed of wave, L is length of resonant cavity and n is the resonant mode.

3 Objectives

I To observe the absorption spectrum of the diode pump laser

II To determine the laser characteristics of diode pump laser

III To create the Diode Pumped Nd:YAG Laser

IV To determine the Laser characteristic of Nd:YAG Laser

V To observe the phenomenon of spiking

VI To observe the generation of the second harmonic using the frequency doubler KTP crystal

VII To observe the TEM modes of the second harmonic and determine their intensity profile

4 Theory

4.1 Types Of Lasers

Lasers are classified into four types based on the type of laser medium used

- Solid State Laser: A solid-state laser is a laser that uses solid as a laser medium. In these lasers, glass or crystalline materials are used.

  Ions are introduced as impurities into host material which can be a glass or crystalline. The process of adding impurities to the substance is called doping. Rare earth elements such as cerium (Ce), erbium (Eu), terbium (Tb) are most commonly used as dopants.

  Materials such as sapphire (Al2O3), neodymium-doped yttrium aluminum garnet (Nd:YAG), Neodymium-doped glass (Nd:glass) and ytterbium-doped glass are used as host materials for laser medium. Out of these, neodymium-doped yttrium aluminum garnet (Nd:YAG) is most commonly used.

- Gas Laser: A gas laser is a laser in which an electric current is discharged through a gas inside the laser medium to produce laser light. In gas lasers, the laser medium is in the gaseous state. Gas lasers are of different types: they are, Helium (He) – Neon (Ne) lasers, argon ion lasers, carbon dioxide lasers (CO2 lasers), carbon monoxide lasers (CO lasers), nitrogen lasers, hydrogen lasers, etc. The type of gas used to construct the laser medium can determine the lasers wavelength or efficiency.

- Liquid Laser: A dye laser is an example of the liquid laser. A dye laser is a laser that uses an organic dye (liquid solution) as the laser medium.

  A dye laser is made up of an organic dye mixed with a solvent. These lasers generate laser light from the excited energy states of organic dyes dissolved in liquid solvents. It produces laser light beam in the near ultraviolet (UV) to the near infrared (IR) region of the spectrum.

- Semi-conductor Laser: In semiconductor lasers, a p-n junction of a semiconductor diode forms the active medium or laser medium. The optical gain is produced within the semiconductor material.

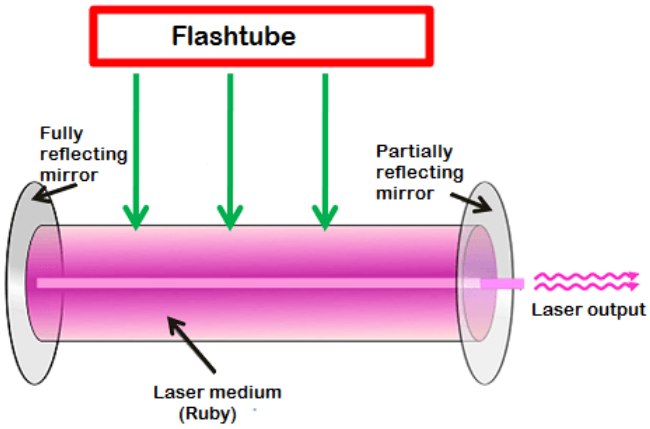
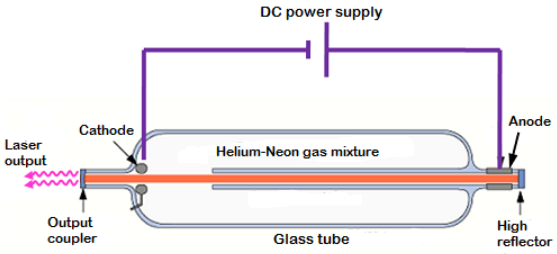
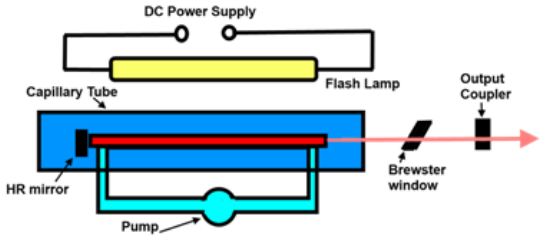


Figure 3: Solid-State Laser[4]



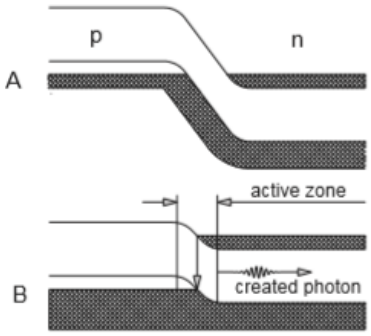
(a) Gas Laser[4]



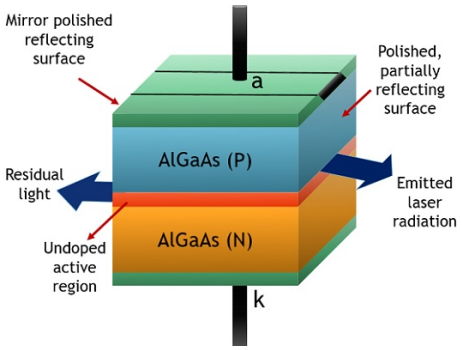
(b) Liquid Laser[5]

4.2 Diode Pump Laser

The pump light source is chosen to be a diode laser because they offer high efficiency and are quite compact in size when compared to the high powered discharge lamps. The active medium is basically a p-n junction.

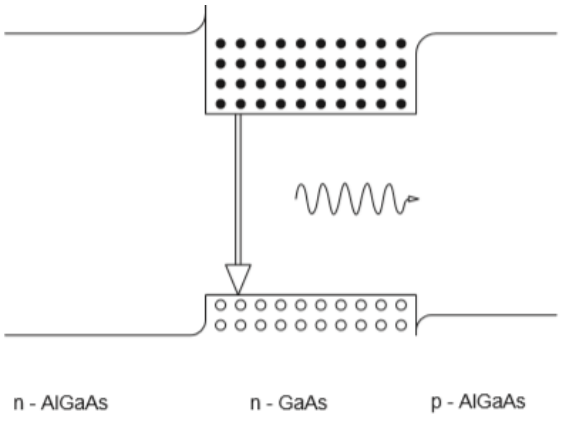


(a) (A)- p-n junction without applied voltage and (B)- p-n junction with applied forward voltage[1]

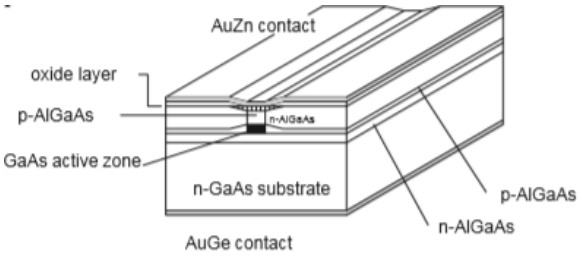


(b) Constructional Detail Of Laser Diode, courtesy Google Images

In 1960's, a threshold current of 100 kA/cm^2 was needed. To reduce the threshold current, something called a double buried hetero-structure is used. In this concept a layer of p conducting GaAlAs is brought on the p layer of the pn transition of GaAs. The slightly higher band gap of GaAlAs compared to GaAs ensures that a potential barrier is created between both materials in a way that charge carriers accumulate here and the formation of inversion is increased and the laser threshold is remarkably lowered. A similar layer is also attached to the n-side of the crystal to further lower the threshold.



(a) Energy Band Gap diagram of a np- double hetero structure[2]



(b) Buried hetero-structure-Active zone has been buried between layers which ensure optical beam guidance[2]

Not only the beam guidance but also the size of the laser mirrors influences the beam geometry. Generally for conventional lasers the mirrors are very large compared with the beam diameter. The laser mirror (crystal gap area of the active zone) of the laser diodes have a very small size, through which the laser beam has to squeeze itself. Diffraction effects will be the consequence and lead to elliptical beam profiles.

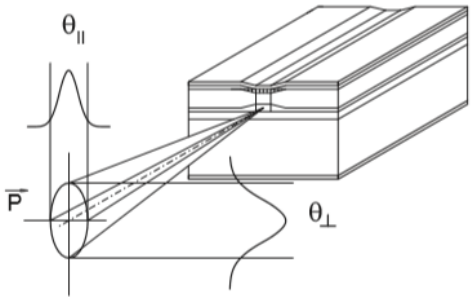


Figure 7: Elliptical Beam Profile OF A Diffraction Limited Laser[1]

Modern diodes are mostly index guided diodes. This means that the laser beam is forced not to leave the resonator laterally by attaching lateral layers of higher refractive index to the active zone. At the gain guided diodes the current is forced to pass along an extremely small path. In this way the direction of the amplification (which is proportional to the current flux) and the laser radiation are determined. At the gain guided diodes the formation of curved wave fronts within the resonator is

disadvantageous since they simulate spherical mirrors. In this case higher injection currents provoke transversal modes which will not appear in index guided diodes because of the plane wave fronts. Laser diodes with intensity profiles following a Gaussian curve and a beam profile which is only limited by diffraction are called Diffraction Limited Lasers (DFL). This is what we have used for our experiment. The wavelength produced by the diode laser is around 808 nm which varies with the injection current and the temperature applied to the laser diode. This variation can be seen using the absorption characteristics of the diode laser.

4.3 Nd:YAG Laser

Neodynium doped Yttrium Aluminium Garnet (Nd:YAG) is a solid state laser in which Nd;YAG is used as a laser gain medium. Nd-YAG is a 4 level laser system.

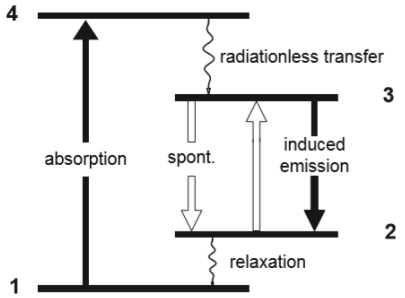


Figure 8: Working Principle Of A 4-Level Laser[1]

Energy Source:- The energy source supplies energy to the active medium to achieve population inversion. Laser diode is being used for this purpose.

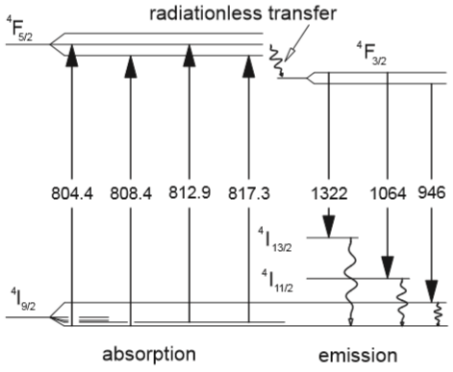


Figure 9: Relevant Energy Levels Of Nd:YAG For Optical Pumping With 808nm Laser Diode[1]

Active Medium:- Synthetic Crystalline material (Yttrium Aluminium Garnet) doped with Nd (neodynium) is the active medium. The lower energy state e^- of the neodynium ions are excited to higher energy state to provide lasing action in the active medium.

Optical Resonator:- The optical resonator used for the amplification is the hemispherical resonator. Here, Nd:YAG crystal is placed between two mirrors one of which is plane and the other is spherical. The plane mirror along with its properties are specified in Module D while the properties of the spherical mirror are already explained in Module E.
The optical resonator is optically stable if after any number of reflections, the light still remains in the resonator due to the imaging characteristics of the mirrors used and does not leave the resonator by protruding beyond the edges of the mirrors. The stability criterion is defined using the inequality:

$$0 \leq g_1.g_2 \leq 1 \quad where\ g_i\ = 1 - \frac{L}{R_i} \tag{4.3.1}$$

Here L is the length of resonator and R is the radius of curvature of the mirror. In our experiment $R_1 = \infty\ and\ R_2 = 100\ mm$ which by stability criterion gives $L \leq 100\ mm$.

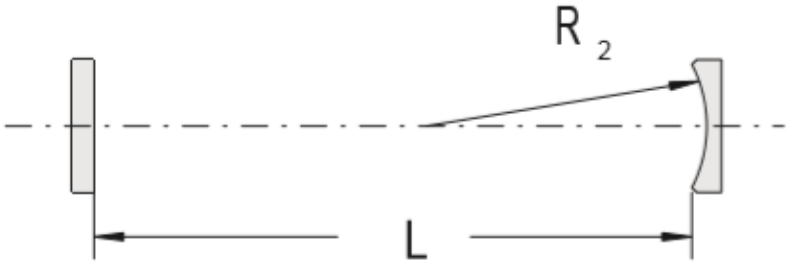


Figure 10: Hemispherical Resonator[2]

Basic Working: Consider a Nd:YAG laser consisting of 4 energy levels with $E_1 \leq E_2 \leq E_3 \leq E_4$. Here E_1 is the lower energy state, say ground state, E_2 is the next higher energy state (first excited state), E_3 is the meta-stable state and E_4 is the pump state where e^- go after getting excited from the pump laser.
Now diode laser supplies energy to the active medium and sends the E_1 e^-'s into the E_4 energy state. Since $\tau_{E_4} \ll \tau_{E_3}$, electrons fall into the E_3 state in a short time but don't leave as quickly. This increases the number of electrons in the E_3 state and population inversion is achieved. After sometime, the e^- of the meta-stable state E_3 release photons by spontaneous emission to fall in to the next lower energy state E_2. Now $\tau_{E_2} \ll \tau_{E_1}$, and e^- jump from E_2 to E_1 by radiation-less transfer. Photons emitted via spontaneous emission interact with e^- of the other meta-stable states to cause stimulated emission creating two coherent photons (one extra photon for every stimulated emission). Now these two photons interact with the active medium to produce 4 coherent photons and the process continues until millions and millions of coherent photons are generated and optical gain is completed.

4.4 Second Harmonic Generation (SHG)[6]

The relation of nonlinear polarization ($\mathbf{P}$) with the electric field ($\mathbf{E}$) can be expressed in a vector form as follows:

$$\mathbf{P} = \epsilon_0\chi\mathbf{E} + \epsilon_0\chi^{(2)}\mathbf{E}.\mathbf{E} + \epsilon_0\chi^{(3)}\mathbf{E}.\mathbf{E}.\mathbf{E} + ... \tag{4.4.1}$$

The second-order susceptibility ($\chi^{(2)}$) is the source of the second-order nonlinearities, such as the second harmonic generation (SHG) i.e., frequency doubling, up and down conversion of wavelengths, parametric amplification and the linear electro-optic effect (LEO). While interacting with a material, if a second order frequency is generated, the total electric field can be expressed as,

$$\tilde{E} = E_1 e^{-i\omega_1 t} + E_2 e^{-i\omega_2 t} + E_1^* e^{i\omega_1 t} + E_2^* e^{i\omega_2 t} \tag{4.4.2}$$

Considering the second order term in scalar form $P = \epsilon_0 \chi^{(2)} E^2$ and putting the electric field of the above form, we get the following terms for the second order optical effects:

$$P(2\omega_1) = \chi^{(2)} E_1^2 \tag{4.4.3}$$

$$P(2\omega_2) = \chi^{(2)} E_2^2 \tag{4.4.4}$$

$$P(\omega_1 + \omega_2) = 2\chi^{(2)} E_1 E_2 \tag{4.4.5}$$

$$P(\omega_1 - \omega_2) = 2\chi^{(2)} E_1 E_2^* \tag{4.4.6}$$

$$P(0) = 2\chi^{(2)} E_1^2 [E_1 E_1^* + E_2 E_2^*] \tag{4.4.7}$$

Above, Eq.4.4.3 and 4.4.4 refer to second harmonic generation (SHG), Eq.4.4.5, the sum frequency generation (SFG), Eq.4.4.6, difference frequency generation (DFG) and Eq.4.4.7, is called the optical rectification.

In general, there are three requirements that are to be met in order for any nonlinear phenomenon to occur, namely, the presence of intense light, conservation of energy and conservation of momentum. In SHG, the two photons of same energy combine to produce a single photon of energy equal to the sum of the energies of both the input photons. In a nutshell, SHG essentially is the welding of two input photons. Assuming the frequencies of the fundamental and second harmonic photons to be $\nu_1 and \nu_2$ respectively and applying the laws of conservation of energy and momentum gives:

$$h\nu_2 = h(2\nu_1) \tag{4.4.8}$$

$$\mathbf{k}_2 = \mathbf{k}_1 + \mathbf{k}_1 \tag{4.4.9}$$

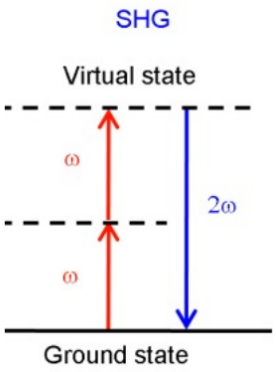
For collinear interaction, the above equation reduces to give the phase matching requirement(here η is the index of refraction)

$$\eta(\nu_2) = \eta(\nu_1) \tag{4.4.10}$$

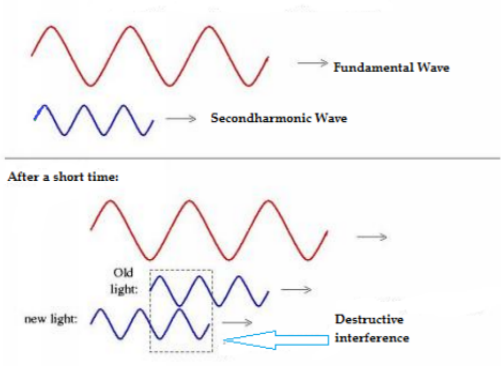
If the index of refraction is same for both the frequencies then the conversion occurs at the highest degree of efficiency. The need for phase matching can be understood as follows. Due to dispersion in the medium the refractive index is different for the fundamental and the second harmonic waves. The result is that the second harmonic wave generated at one location is not interfering constructively with the second harmonic wave generated in the crystal at some other location at a later time. This greatly reduces the intensity of the second harmonic wave since the waves may interfere destructively too.

Once the phase matching is ensured, the power of the second harmonic follows the relation:

$$P_{2\nu} \propto P_\nu^2 \tag{4.4.11}$$



(a) Two photons of frequency ν combining to give one of frequency 2ν



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(b) SHG without phase matching. Figure courtesy: Anne Marie March, SUNY Stony Brook Optics Rotation Project 4, Spring 2001, "Frequency Doubling Using LBO in a Resonant External Enhancement Cavity"

4.5 Spiking Phenomenon

The spiking phenomenon exploits the non-linear differential equations for photon density and population inversion when the laser is not in the steady state. At steady state both the rate of change of photon field and population inversion is 0 while it has a non-zero value at a non-steady state. A large deviation from the steady state undoubtedly occurs when the laser is switched on or when the pump-light source is switched on. Until the threshold pump power P_{th} is reached, there are practically no photons present in the resonator.

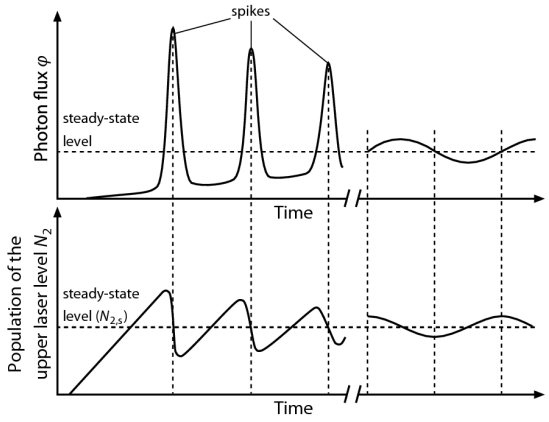


Figure 12: Graphs for photon flux and Population Of Metastable State for a Pulsed Laser Operation, Taken from: Eichler H.J., Eichler J., Lux O. (2018) Pulsed Operation. In: Lasers. Springer Series in Optical Sciences, vol 220. Springer, Cham

When the population inversion reaches the threshold, a photon field is formed. However, due to the resonator propagation time, it takes a while until the photon density reaches the steady state value. During this period the inversion, which rises linearly with time, increases above the value of

the threshold inversion. This in turn means a more rapid increase in the photon density. This rise is so rapid that the inversion falls to a value slightly below the threshold and the laser oscillation stops (Fig. 1.8). The process starts again, but this time the laser is only slightly below the threshold and the expected inversion overshoot is not so large as before. In this manner the system approaches the steady state.

4.6 Transverse Electromagnetic Modes

Modes are the standing oscillating electromagnetic waves which are defined by the cavity geometry. A transverse mode of electromagnetic radiation is a particular electromagnetic field pattern of the radiation in the plane perpendicular (i.e., transverse) to the radiation's propagation direction. Transverse modes occur in radio waves and microwaves confined to a waveguide, and also in light waves in an optical fiber and in a laser's optical resonator.

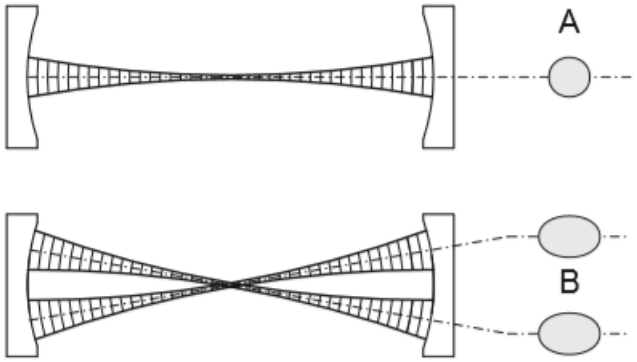


Figure 13: A Spherical Resonator with oscillation in (A) Fundamental TEM_{00} mode and (B) $\text{TEM}_0 1$ Mode

The boundary conditions for an EM wave at the boundary of a waveguide are:

$$B_{\perp inside} = B_{\perp outside} \tag{4.6.1}$$

$$E_{\parallel inside} = E_{\parallel outside} \tag{4.6.2}$$

In a TEM_{Amn} (Transverse Electro-Magnetic Mode), in addition to the boundary conditions, no electric or magnetic field exists in the direction of propagation of the laser. Further the indices m,n are integers which tell give the number of intensity spots minus 1 in the x and y axes respectively.
In a laser with cylindrical symmetry, the transverse mode patterns are described by a combination of a Gaussian beam profile with a Laguerre polynomial. The modes are denoted TEM_{mn} where m and n are integers labeling the radial and angular mode orders, respectively. With m = n = 0, the TEM_{00} mode is the lowest order. It is the fundamental transverse mode of the laser resonator and has the same form as a Gaussian beam. The pattern has a single lobe, and has a constant phase across the mode. Modes with increasing m show concentric rings of intensity, and modes with increasing n show angularly distributed lobes.

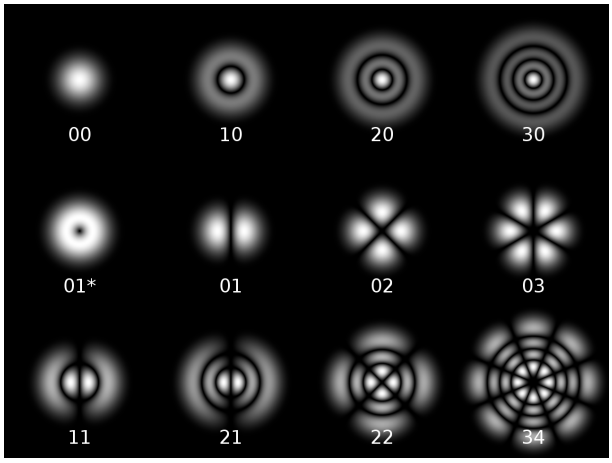


Figure 14: TEM mode pattern for a laser with cylindrical symmetry,By DrBob at English Wikipedia, CC BY-SA 3.0

5 Apparatus Required

- Optical Bench
- Modules A-K with appropriate holders
- Nd-YAG Crystal And KTP Crystal
- ED-0060 Photo-detector Conditioning Box
- Diode Laser Diode And Peltier's Element Controller ED-020
- IR Card which can detect the 1064 radiation
- Optical Power Meter (Ophir PD-600)
- Grating And Adjustable slit
- Screen
- Oscilloscope
- Digital Multi-meter
- Computer System With Appropriate Software
- Connecting Cords

5.1 The Modules[1]

- **Module A - Diode laser 808 nm 500 mW** : This module provides the laser light source consisting of a temperature controlled laser diode. The maximum output power is 500 mW at 20 C. The laser module is fixed into an adjustment holder which is attached to a carrier to place the module onto the optical rail having a length of 500 mm. By means of a Peltier's element the temperature of the laser diode can be changed in a range from 15 to 40 C. To control the injection current as well the temperature the diode laser head is connected to the ED-0020 controller.

- **Module B - Collimator** : The collimator consists of a three-lens system (1) with a short focal length (f=8mm) and a large aperture in order to collimate divergent laser radiation emitting from the laser diode.

- **Module C - Focussing Unit** : This unit is used to focus the collimated diode laser beam into the YAG rod which is mounted in Module D. The lens has a focal length of 60 mm and is mounted into a 25 mm click holder.

- **Module D: Laser mirror adjustment holder with Nd:YAG rod** : Module D and E form the resonator of the Nd-YAG laser. The adjustable holders have the duty of adjusting the relevant resonator mirror so that the common optical axis is aligned perpendicular to the mirrors. The plane-parallel YAG rod, which is 5 mm long and has a diameter of 3 mm, is located in an exchangeable mount in Module D. A coating, which is highly reflective at the laser wavelength of 1064 nm, has been placed onto one end of the rod which also forms the left resonator mirror. The coating is designed in such a way that the maximum pump-light radiation can penetrate the highly reflective layer with only 20% losses. The other end of the rod is coated with a high-quality anti-reflection layer for 1064 nm in order to keep the internal resonator losses as low as possible. In addition the back side of the YAG rod is coated with a high reflective layer for 532 nm in order to redirect the green light to the output of the resonator.

- **Module E: Laser mirror adjustment holder** : This module contains the second resonator mirror. Also as with the laser rod, it can be screwed out with its fixture from the holder. The mirror has a diameter of 1/2 inch and a radius of curvature of 100 mm. This mirror has a high reflectivity (¿99.98%) to keep as much of fundamental power within the resonator.

- **Module F: Filter plate holder** : The filter plate holder comes with two filters. The colour filter RG1000 suppresses the pumping radiation of 808 nm. The second filter BG39 allows only the radiation of 532 nm to pass.

- **Module G: SiPIN Photodiode** : This unit consists of a SiPIN photodiode accommodated in a 25 mm housing with attached cable. The detector is connected to the signal conditioner box.

- **Module K - Frequency Doubler** : The second harmonic at 532 nm of the fundamental wave at 1064 nm is generated by means of a KTP crystal. The crystal has a size of 2x2x6 mm and is fitted into a special holder to allow the cleaning from time to time. The fitting is mounted into a holder which is inserted into the 5 axes adjustment unit.

6 Experimental Setup[1]

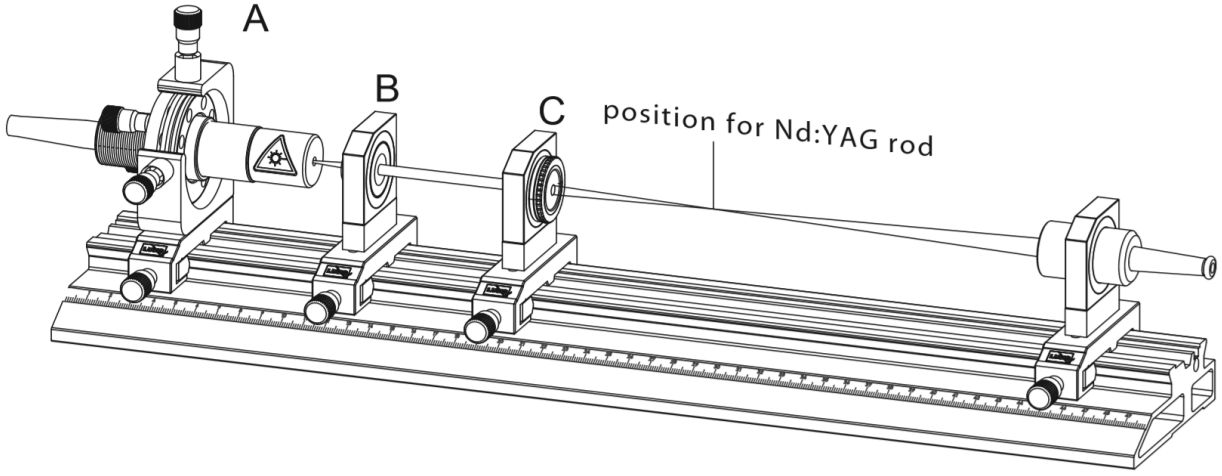


Figure 15: Setup For Finding Diode-Laser Characteristics

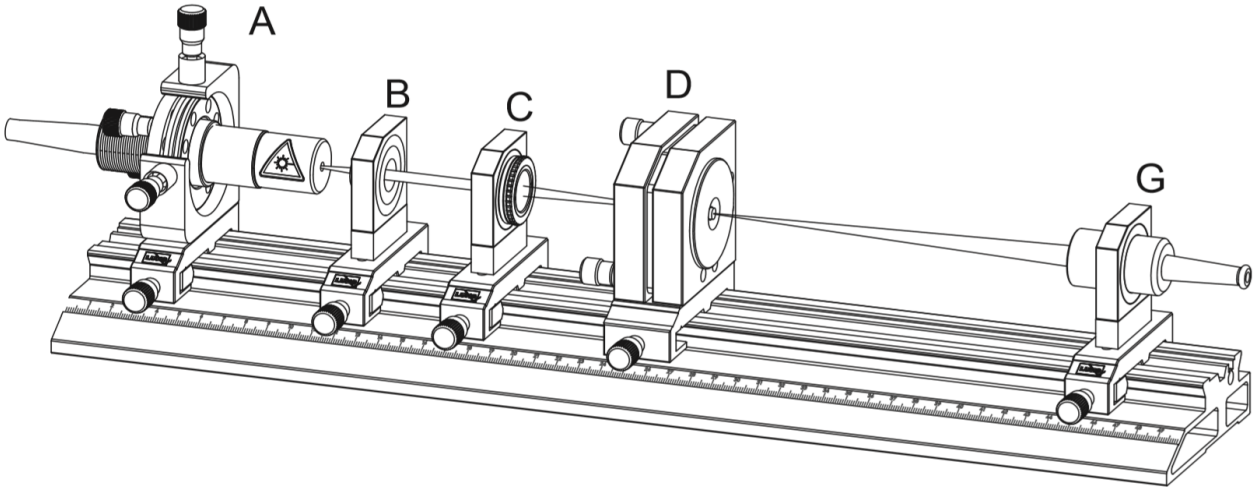


Figure 16: Setup For Finding Absorption Characteristics Of Diode-Laser

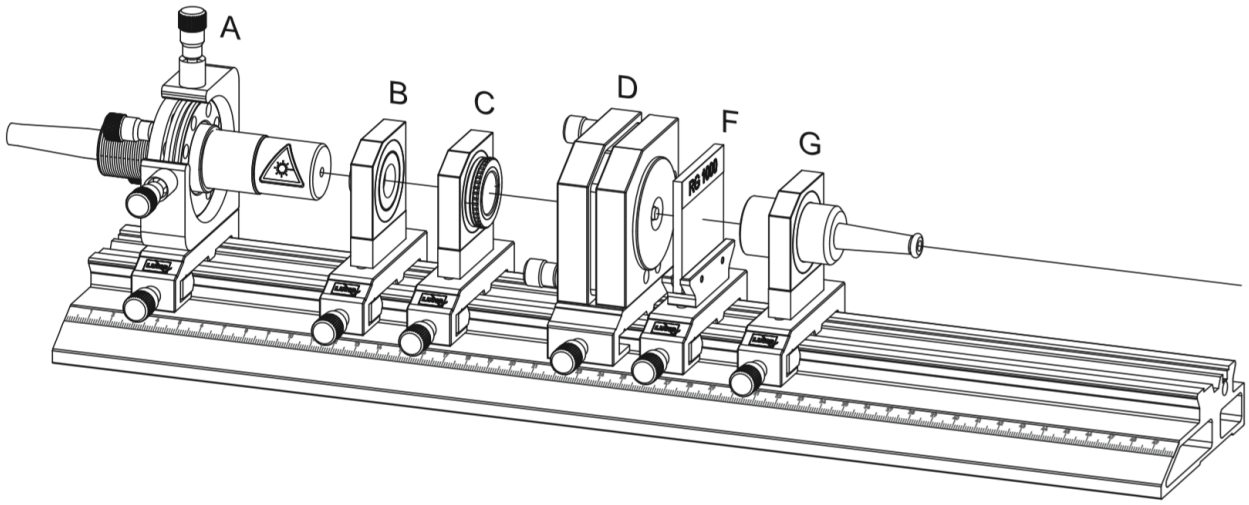


Figure 17: Setup For ${}^4F_{\frac{3}{2}}$ Life Time Measurement

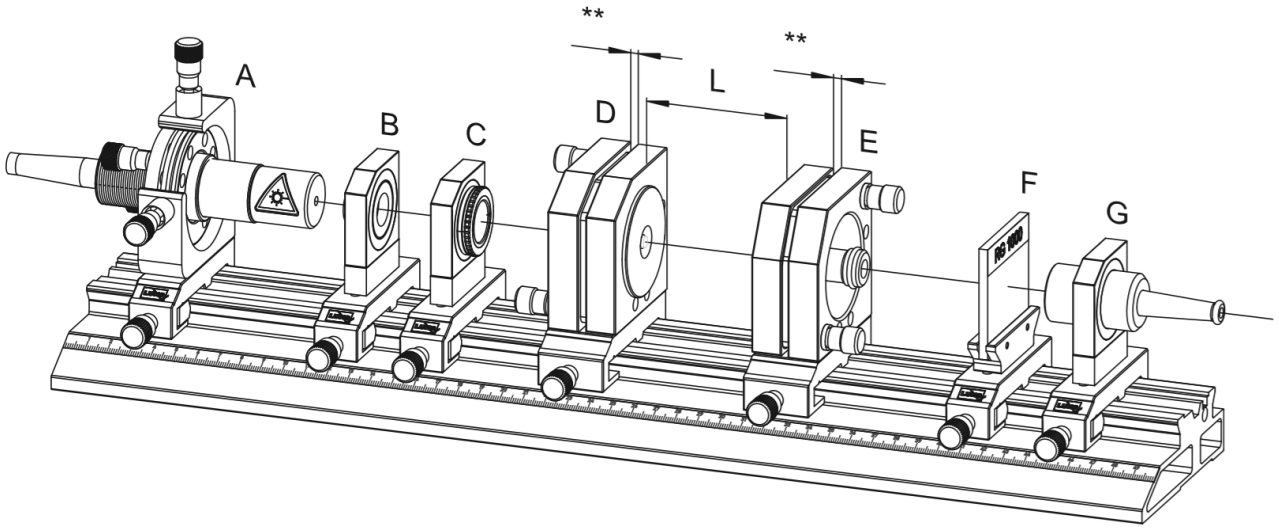


Figure 18: Nd-YAG LASER Setup

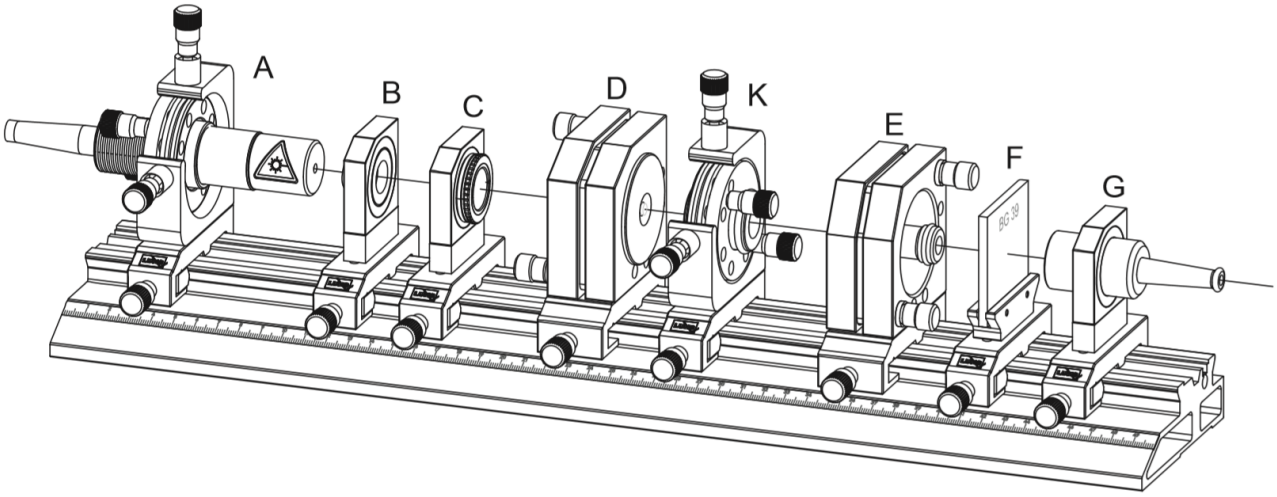


Figure 19: Setup For Second Harmonic Generation

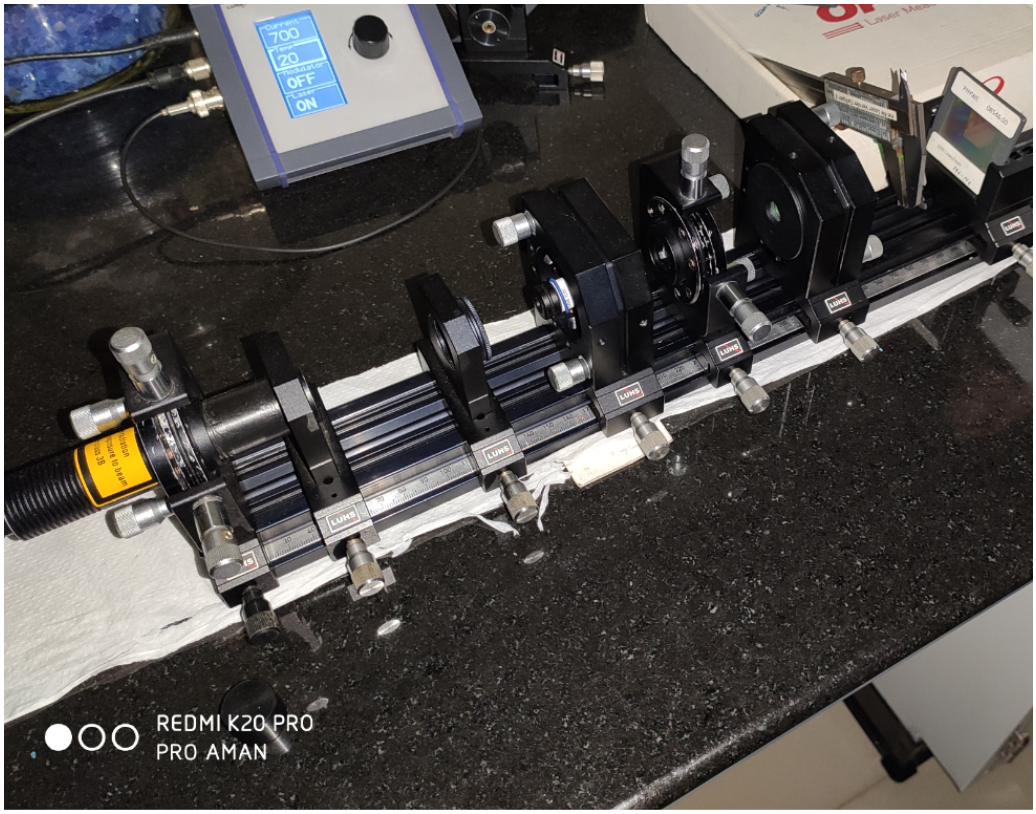


Figure 20: Setup To Observe Diffraction In Second Harmonic

7 Observations And Calculations

The graphing and analysis is done in the next section. This section simply summarises my basic observations. All the graphs have been plotted in python and the curve fitting and error in fits is done using the curve_fit function of python. Note that all the errors mentioned here are the statistical error. I have only mentioned statistical errors if they are greater than 1%. Otherwise they donot make any sense since the propagational error which arises due to the least count of our instrument is in the range of 1-3%.

* The absorption spectrum is of diode pump laser is observed. A maximum absorption efficiency of 0.9 is observed which is one of the reasons why semiconductor diode was being used as a pump source.

* The wavelength of diode laser decreases as temperature increases at constant power in the working region of the laser. The parameters share a linear dependence.

* The diode laser has a threshold current of 144.004 $\pm$ 1.756 mA below which no significant laser power is observed. The output power is function of both temperature and the injection current of the diode laser.

* The production of Nd-YAG laser requires extremely fine adjustment of each and every single component especially the resonator length so that 1064 nm wavelength is one of the resonant modes of the cavity.

* The Nd-YAG laser formed has a threshold current of 140.426 $\pm$ 4.218 mA. The relationship between diode temperature and ouput Nd-YAG power and hence output wavelength is quite erratic and doesn't follow a general trend.

* Spiking phenomenon, wherein the output power spikes up to ten or even hundred fold times is observed. Further, a time lag between turning on the laser and getting the first spike is observed.

* The non-linear optical effect of second harmonic generation is observed and a light wave of 532 nm is produced using the fundamental laser light of 1064 nm. The fundamental wave lies in the infrared region while the second harmonic is in the visible range perceived as green color.

* The second harmonic is only generated when the KTP crystal is kept inside the resonant cavity. If the KTP crystal is outside the resonant cavity no green light is observed.

* The power of the second harmonic is observed to be proportional to the power of the fundamental wave.

* A variety of TEM modes for the second harmonic are observed. The TEM_{00} mode looks like a bright spot on the screen while TEM_{10} mode looks like a double lobed structure much like the p_x orbital on the x-y plane. As the order of modes increases, the number of nodes increases both along the x and the y axes.

* The second harmonic is passed through a diffraction grating with 1200 lines per mm. In second year, we used the same grating to separate sodium d-lines which are just 0.6 nm apart, However,

no such separate lines were seen with the second harmonic green light. This simply implies that the linewidth of the green laser light and hence our original laser is strictly less than 0.6 nm.

Figure 21: Diffraction Of Second Harmonic Using 1200 lines/mm grating

7.1 $^4F_{\frac{3}{2}}$ Mean Life Time Measurement

Assuming that the electron follows first order decay statistics:

$$V = V_0 e^{-t/\tau} \tag{7.1.1}$$

when t=τ, V=V_0/e and we have used this voltage to determine the mean life time of the meta-stable state from the oscilloscope directly. The experiment was done at the gain resistance of 10kΩ where the mean response time of the detector is 10 μs.
The experiment is repeated 10 times to get the following mean life times in μs:
250,260,265,280,275,250,265,260,255,275. Hence:

$$\tau^{avg}_{4F_{\frac{3}{2}}} = 263.5\mu s$$
$$\Delta\tau^{std}_{4F_{\frac{3}{2}}} = 10.5\mu s$$

In addition to the standard deviation there is an additional uncertainty of 10 μs that arises due to response time of the detector. However since the measurements are nearly two orders apart, the response time of the detector doesn't really matter. However for higher gain resistances the response time of the detector is comparable to mean life time and we are unable to differentiate whether the output is the response of the detector itself or the electrons in the meta-stable state.

8 Data Analysis And Graphs

8.1 Absorption Spectrum Of Pump Diode LASER

In the following experiment, the dependence of the wavelength of the diode laser beam on the diode temperature is being explored. We used the well-known absorption lines of the Nd-YAG crystal to perform this experiment. According to the energy level diagram of the Neodynium ions, there are four absorption transitions which correspond to 804.4 nm ,808.4 nm ,812.9 nm and 817.4 nm. In our experiment we were only able to see one of these four transitions due to limitations in our setup. Since, we were only able to vary the temperature in a range of 15^o - 35^o, we were only able to retrieve a part of the spectrum. In the above graph, the lower the output voltage higher is the absorption. This can

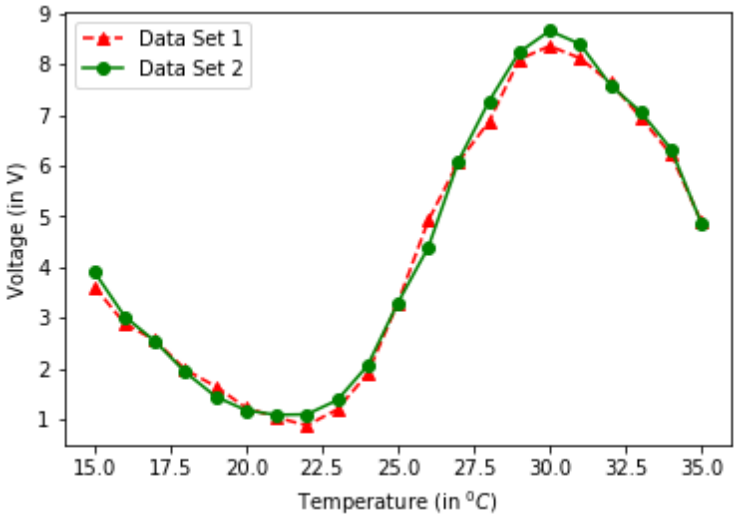


Figure 22: Diode Laser Output Voltage vs Temperature

be seen from the formula:

$$a = \frac{V(T) - V_{max}}{V_{max}} * 100 \tag{8.1.1}$$

where V(T) is the voltage at temperature T with Nd-YAG crystal, V_{max} is the maximum output voltage without any Nd-YAG crystal. Note that while taking these readings the injection current was high enough so that the change in the maximum output voltage is negligible over the variation of temperature. The absorption graph is plotted in Fig 2.

From our graph it cannot be conclusively interpreted to which wavelength does our peak correspond to. However, results from other experiments tell us that this peak should correspond to 808.4 nm since the 804.4 nm peak is expected to come before 15^oC and the peak in this range correspond to the former.

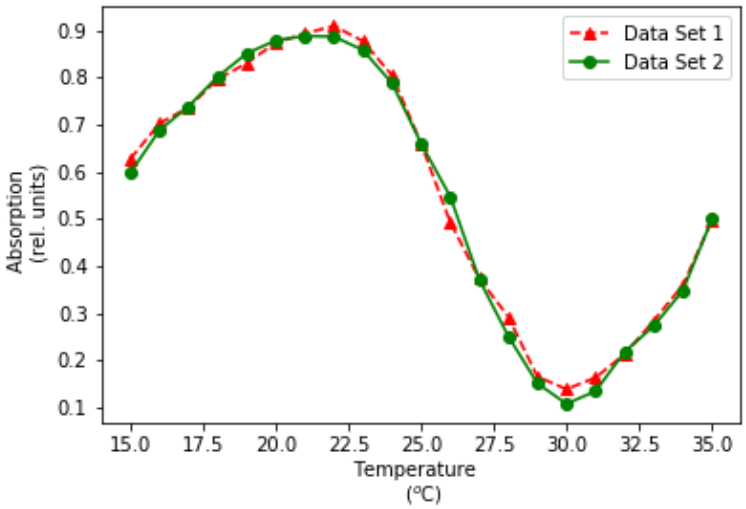


Figure 23: Absorption Measurements For Nd-YAG vs LASER Diode Temperature

8.2 Wavelength And Temperature Dependence Of Diode LASER

Since the change in injection current not only changes the output power of the diode LASER but also the the wavelength of the outgoing light. The variation however small will effect the formation of the Nd-YAG LASER as the absorption efficiency for our crystal is maximum at particular wavelengths. Hence , it is necessary to be able to vary the pump power without leaving the absorption peak. The graph of injection current vs Temperature fits a straight line curve with a negative slope. This tells us that an increase in injection current increases the wavelength of the diode LASER and hence temperature has to be reduced to compensate and remain at the same wavelength.

```
For A x +C, respective coefficients are:
[ -38.76373627 1089.80769246]

<matplotlib.legend.Legend at 0xb5ebf60>
```

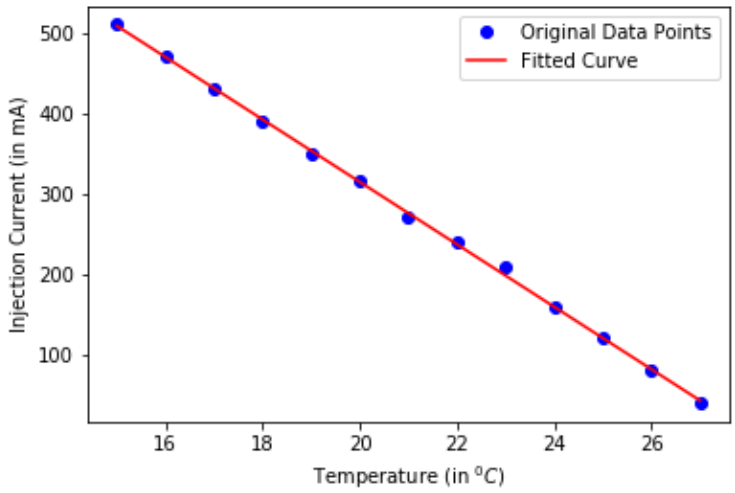


Figure 24: LASER Diode characteristic for operation at 808.4 nm

8.3 LASER Diode Output Power

The power of the diode LASER was measured using an optical power meter. The aim was to discover some relation between the output power of diode LASER and injection current.

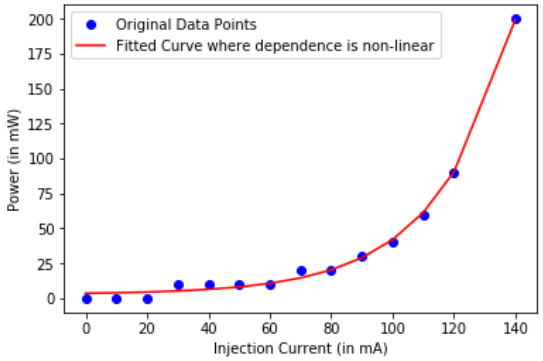


Figure 25: Diode LASER Power characteristic Below Threshold Current at 20^oC

From Fig 26, there can clearly be seen a threshold current after which the diode LASER starts working. After achieving threshold current, the graph fits very well with a straight line. The threshold current for our pump LASER comes out to be 144.004mA.
One thing to notice is that the power of the pump LASER doesn't go exactly to 0 below threshold

current. Fig 25 gives a zoomed version of that part of the graph which fits greatly with the equation of proportional rate growth of increase (non-linear fit).

```
For A x +C, respective coefficients are:
[  0.14604814 -21.0316356 ]
144.0048142708559
```

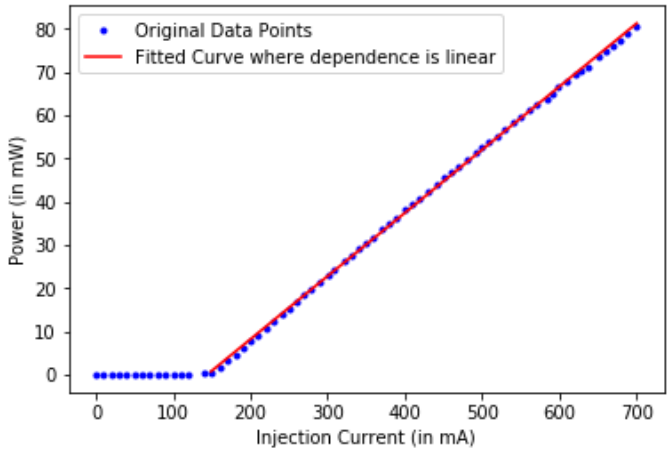


Figure 26: Diode LASER Power characteristic at 20^oC

The power output also depends on the temperature of the laser diode. The current at which the two curves first separate is the threshold current. Below this point only spontaneous emission was responsible for lasing and hence almost no laser was observed. After the threshold is achieved, the optical pump had enough power to create a population inversion which lead to stimulated emission and hence the lasing action.

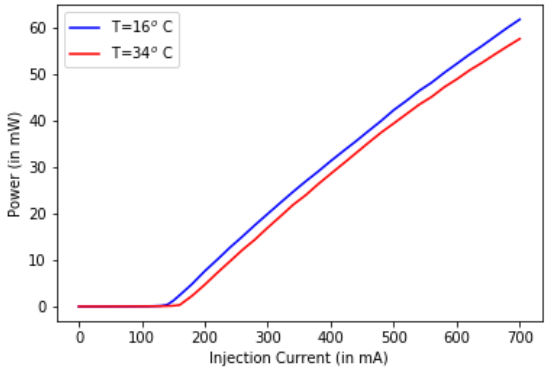


Figure 27: Dependence of Laser Power On Injection Current With Temperature As A Parameter

8.4 Measurement Of Threshold And Output Power Of Nd-YAG LASER

The power characteristic for the Nd-YAG LASER comes out to be 140. Just like for the pumped diode laser, the graph possesses a threshold current. A clear difference that was observed is the saturation of output power near the range of 600-700 mA injection current, unlike that of diode pump laser which kept on increasing linearly after the threshold current. The graph fits with a straight line only in the middle part which we have identified as the working region of the Nd-YAG LASER.

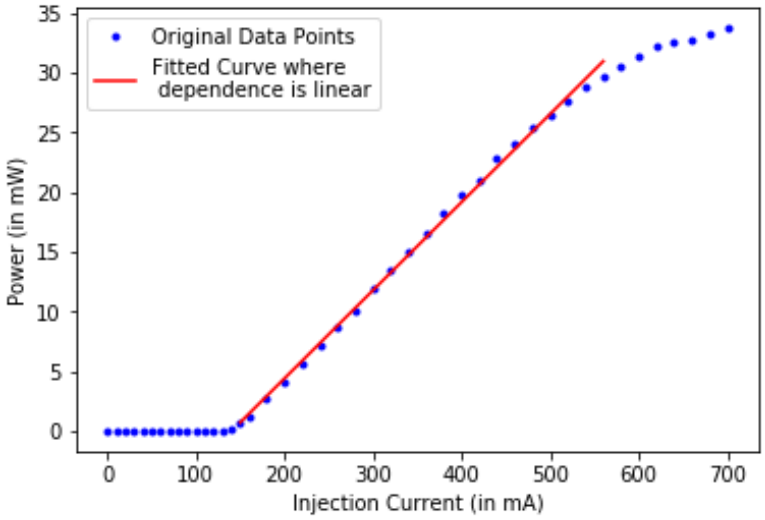


Figure 28: Nd-YAG LASER Power characteristic at 20^oC

```
For A x +C, respective coefficients are:
[  0.0737991  -10.36332241]
140.42612749288804
```

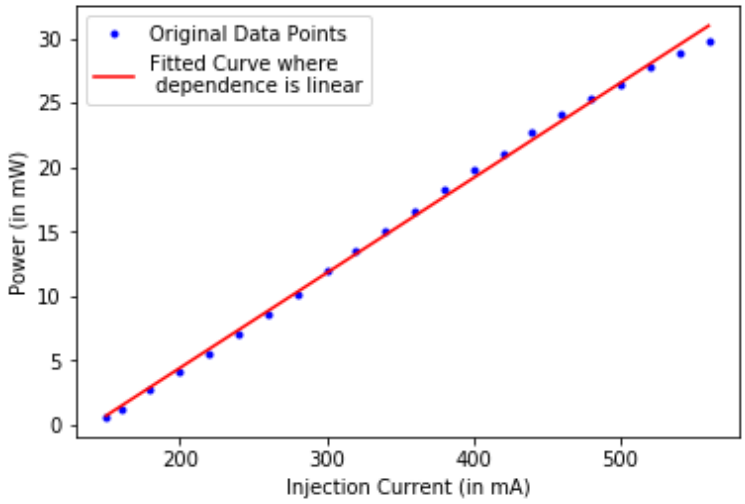


Figure 29: Straight Line Fit In The Working Region Of Nd-YAG

8.5 Nd-YAG Output Power At Various Pump Wavelengths

The relationship of output power with temperature (and hence changing wavelengths) at constant injection current seems to be erratic and don't show any general behaviour. The special thing of the graphs is the presence of a broad peak. This shows that the output power peaks up when the wavelength of the diode LASER is tuned to the absorption peaks oif the Nd-YAG crystal.

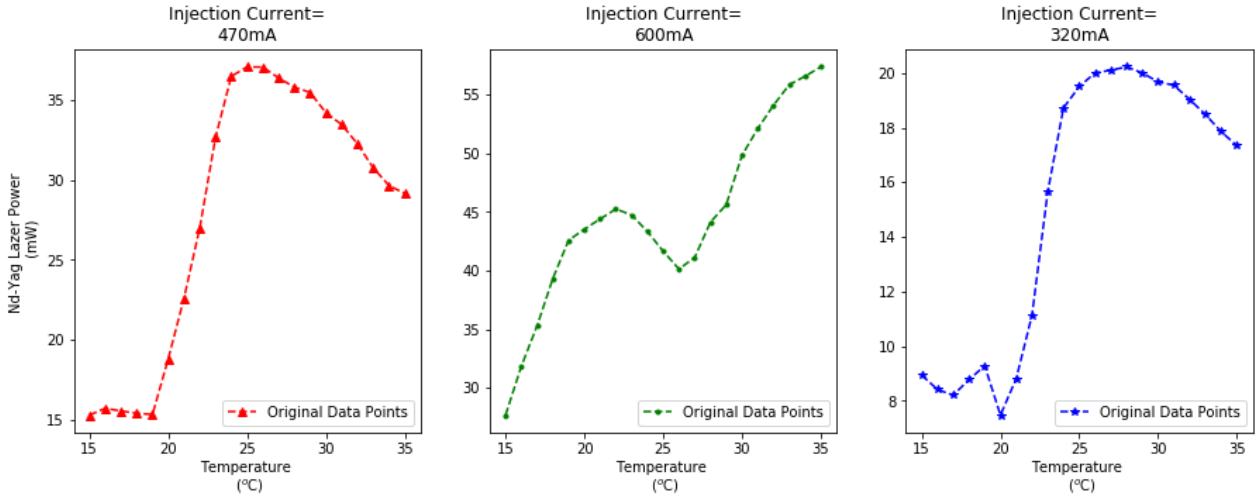


Figure 30: Nd-YAG LASER Power vs Temperature At fixed Injection Current

8.6 Demonstration Of Spiking

Spiking is a phenomenon when the LASER is not at the steady state. Spikes in output voltages which are much higher that the steady state voltage is observed. In our experiment an amplification of almost 100 times than the steady state voltage is observed.

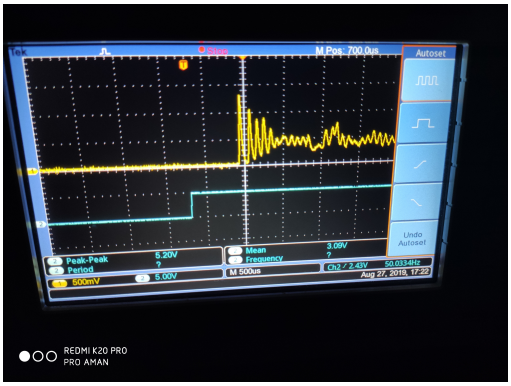
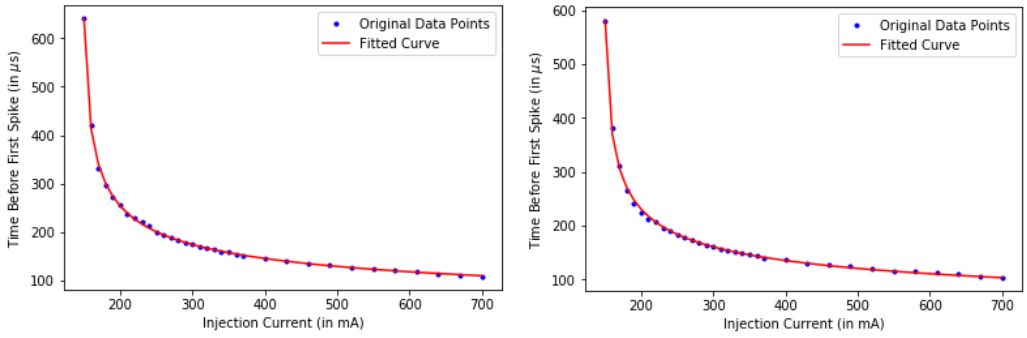


Figure 31: Spiking Phenomenon:Upper trace shows Photo-detector Signal And Lower shows Injection Current

As one can observe form the figure, there is a time lag between the switching on of the diode laser and the overshooting in photon density which leads to the spiking phenomenon. We tried to find out if there was a dependence between the two variables and plotted the curve between the time lag before first spike and the injection current. The graph fitted extremely well for the function $y = A(x - x_0)^B$ with A, x_0 and B as parameters for two sets of readings.



(a) Data Set 1

(b) Data Set 2

Figure 32: Graph Of Time Lag Before First Spike And Injection Current

In the above fitting A is just a scaling factor. However, B must represent the threshold current for Nd:YAG Laser as below that point the laser is just operating mainly via spontaneous emission. The average value obtained is 146.05 mA which is quite close to the threshold value of 140 mA as observed earlier. The average value of c $\approx -1/3$ and the exactness of threshold current help us formulate the

following relation between time lag of first spike and injection current of pump diode laser:

$$\tau_{spike1} \propto \frac{1}{\sqrt[3]{(I - I_{Th})}} \tag{8.6.1}$$

8.7 Power Relationship Between Second Harmonic Wave And Fundamental Wave

To verify the quadratic relationship between the power of second harmonic and the fundamental wave, a graph between the two was plotted and then fitted using the function $A(x\text{-}B)^2$. The result is Fig 31.

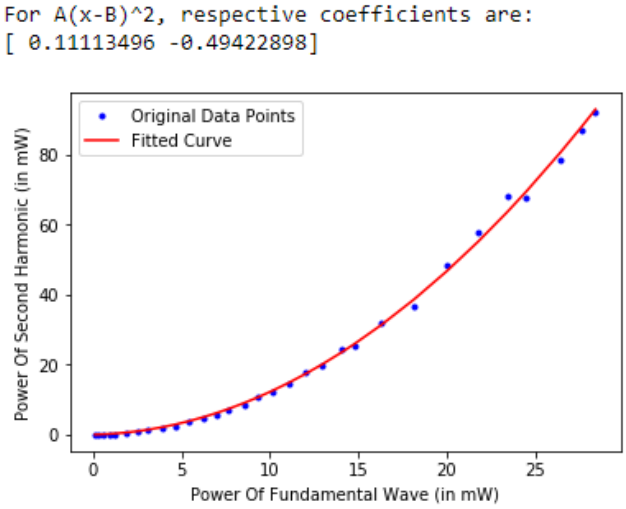


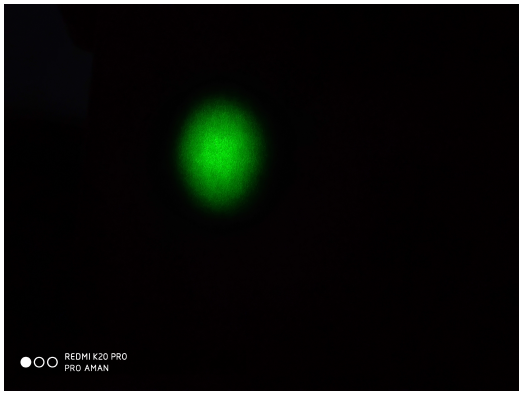
Figure 33: Graph to observe Quadratic Power Relationship Between the second harmonic and the Fundamental Wave

As one can observe, the fitted graph matches with the experimental points quite well. The value of parameter B is quite small and the graph can be thought of as passing through the origin. Further, one can see that the fundamental wave power is quite less than the power of the second harmonic wave. The reason to this is quite simple. The power measurements are taken outside the resonant cavity. Most of the fundamental wave is contained within the resonant cavity by placing highly reflective mirrors sensitive to that wavelength. Hence, the output power measured of the fundamental wave is only a very very small fraction of the actual power within the resonant cavity. However, the resonant cavity doesn't care about the second harmonic and allows it's transmission..

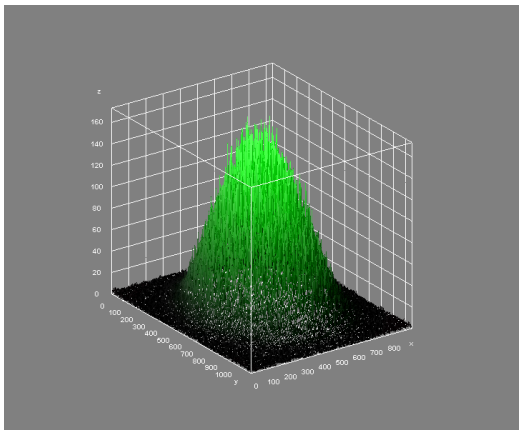
8.8 Observing The TEM Modes Of The Second Harmonic

The TEM Modes observed in this experiment are obtained using a hemispherical resonator.

8.8.1 TEM_{00} Mode



(a) TEM Mode As Observed On A Screen



(b) Intensity Profile Of TEM Mode: z-axis shows intensity while the x-y axes form the plane on which transverse mode is observed

Figure 34: TEM 00 MODE

The TEM_{00} Mode is the most basic and one of the most important modes that can be produced in a laser cavity resonator. This is the only mode which has no radial or angular nodes and appears as a bright spot on the screen. The TEM_{00} mode is supposed to be symmetric hence following a Gaussian intensity profile. This can be intuitively seen in the 3-D figure, which suggests a global maxima somewhere near the centre with continuously decreasing values as one goes away from it. However a more robust analysis of intensity profiling can be done by graphing the intensity of the spot along an axis which passes through the center (brightest point).

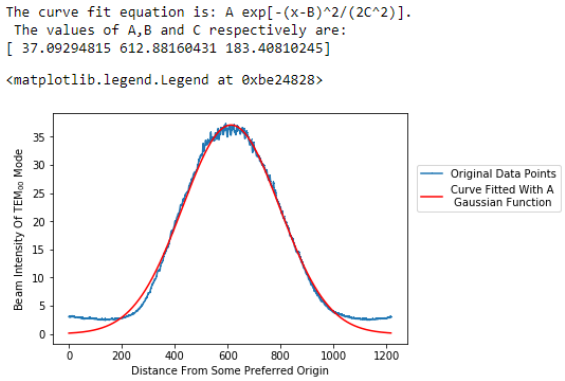
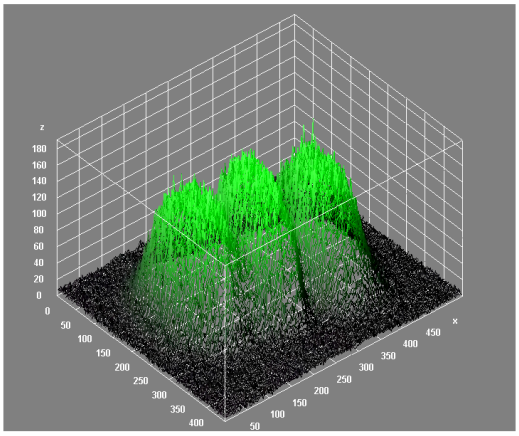


Figure 35: Verifying The Gaussian Profile Of TEM_{00} Mode

As expected, the intensity profile fits very well with a gaussian distribution. The peak is quite broad which is reflected in the value of parameter C.

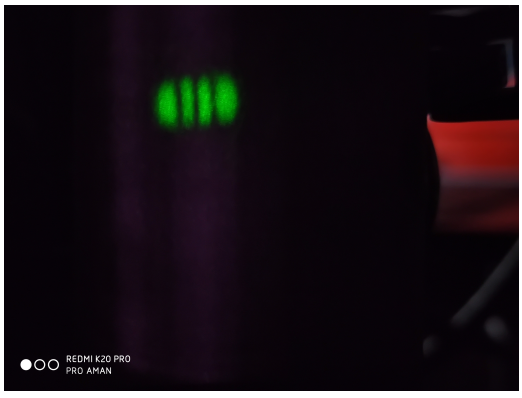
8.8.2 Observation Of Various Other TEM Modes

(a) Mode As Observed On A Screen

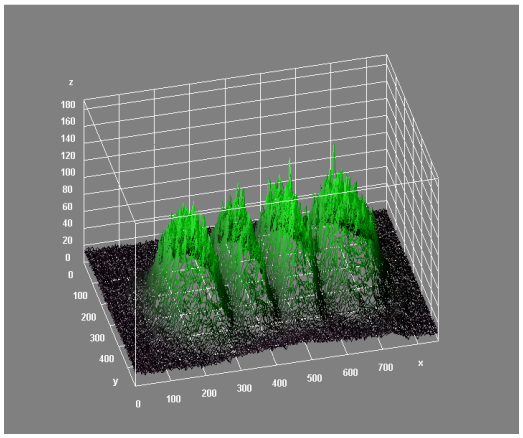


(b) Intensity Profile Of TEM Mode: z-axis shows intensity while the x-y axes form the plane on which transverse mode is observed

Figure 36: TEM 20 MODE

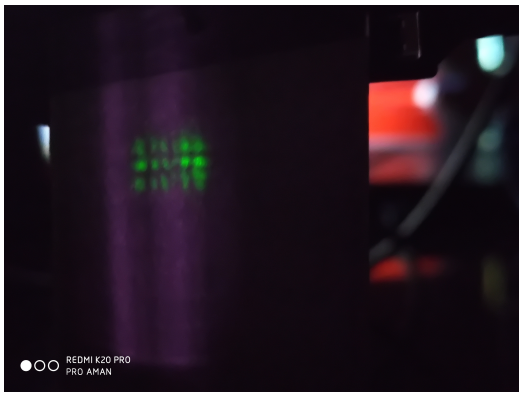


(a) Mode As Observed On A Screen

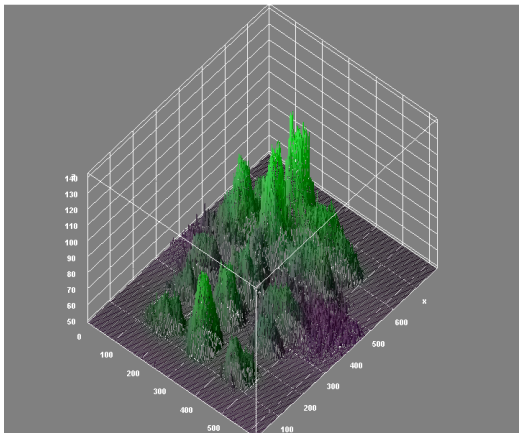


(b) Intensity Profile Of TEM Mode: z-axis shows intensity while the x-y axes form the plane on which transverse mode is observed

Figure 37: TEM 30 MODE

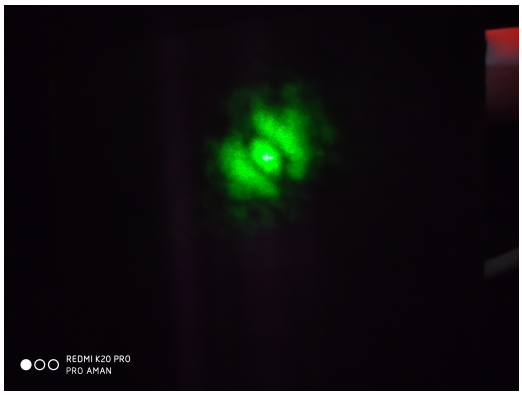


(a) Mode As Observed On A Screen

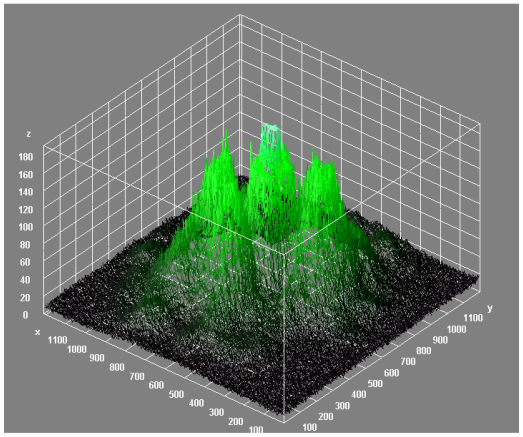


(b) Intensity Profile Of TEM Mode: z-axis shows intensity while the x-y axes form the plane on which transverse mode is observed

Figure 38: TEM 52 MODE

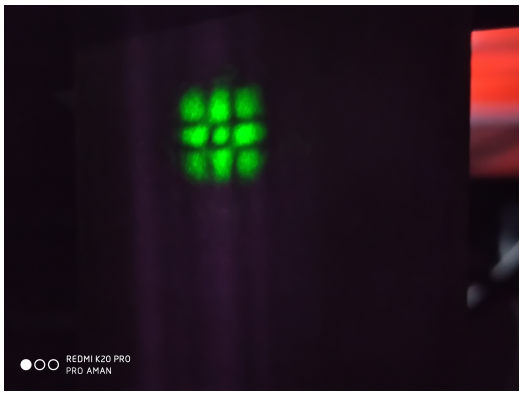


(a) Mode As Observed On A Screen

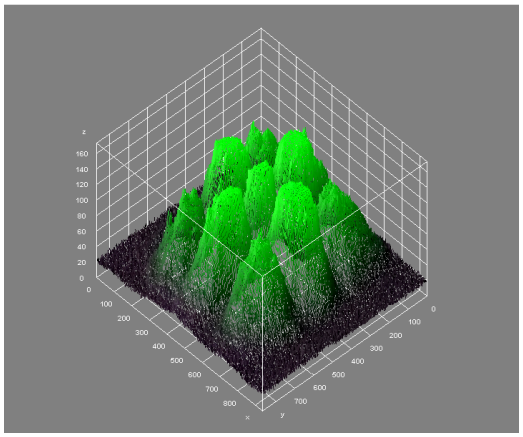


(b) Intensity Profile Of TEM Mode: z-axis shows intensity while the x-y axes form the plane on which transverse mode is observed

Figure 39: TEM 02 MODE

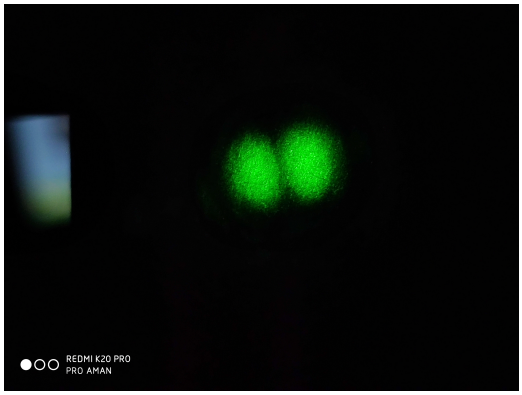


(a) Mode As Observed On A Screen

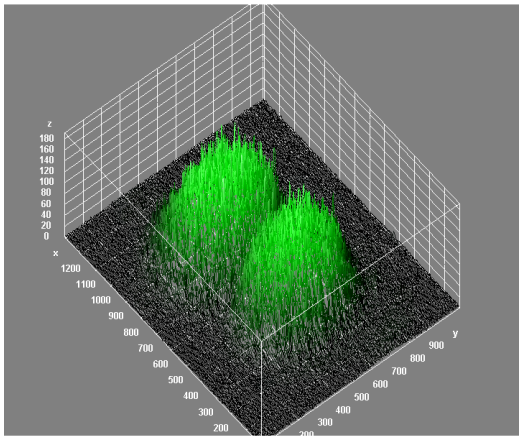


(b) Intensity Profile Of TEM Mode: z-axis shows intensity while the x-y axes form the plane on which transverse mode is observed

Figure 40: TEM 22 MODE



(a) Mode As Observed On A Screen



(b) Intensity Profile Of TEM Mode: z-axis shows intensity while the x-y axes form the plane on which transverse mode is observed

Figure 41: TEM 10 MODE

9 Results And Conclusions

- The absorption spectrum of diode pump laser is observed. The change in absorptivity tells us that the output wavelength of diode laser depends on the temperature applied to it. Else there was no reason to see a change in absorption efficiency for Nd-YAG crystal.
- The diode laser has a threshold current of 144.004 ± 1.756 mA below which no significant laser power is observed. The output power is function of both temperature and the injection current of the diode laser. Below the threshold current only spontaneous emission contributes and because two independently generated photons via spontaneous emission have no reason to be coherent hence almost no laser power is observed.
- The mean life time of the electrons in the meta-stable state ${}^4F_{\frac{3}{2}}$ is found to be 263.5 ± 10.5 μs
- A fully functioning Nd-YAG laser with a threshold current of 140.426 ± 4.218 mA is created.
- Using the Nd-YAG laser in pulsed operation the phenomenon of spiking is observed. The first voltage spike is nearly hundred times the steady state voltage. In addition to this, a time lag is observed between the occurrence of first spike and switching on the laser. Using appropriate analysis, the following experimental relation was concluded:

$$\tau_{spike1} \propto \frac{1}{\sqrt[3]{(I - I_{Th})}}$$

  Where τ_{spike1} is the time lag and I is injection current of diode laser.
- The second harmonic is observed using the frequency doubled KTP crystal. The power relationship between the waves of second and first harmonic is verified.
- The reason that second harmonic is only observed when the KTP crystal is within the resonant cavity is due the fact that this effect is a non-linear effect and only comes into significance if the intensity of the fundamental wave is very high. Since most of the fundamental wave is reflected back into the cavity by the system of mirrors, the intensity outside the resonator cavity is not high enough to activate the non-linear effects and hence show the second harmonic generation.
- A variety of TEM modes are observed for the second harmonic. Out of these I have focused on TEM_{00} mode and found that it follows a gaussian intensity profile in two dimensions which was expected from the theory.

10 Additional Possible Future Experiments

- A circular adjustable iris with proper holder can be provided for better resolution and analysis of the TEM_{00} mode.
- A Passive Q-Switched Laser can be created if a saturable absorber cystal (Cr:YAG) is provided. Similarly a pockel's cell with a rotable Brewster window along with a pockel cell driver can be provided to produce an active Q-Switched Laser.

- Using the Q-Switched Laser both intra-cavity and Extra-cavity second harmonic generation can be performed which in my opinion is a phenomenal feat.

11 Acknowledgements

The success and final outcome of this project required a lot of guidance and assistance from many people and I am extremely privileged to have got this all along the completion of my project. I owe my deep gratitude to our project supervisor Dr Ashok K Mohapatra, who took keen interest on our project work and guided us all along, till the completion of our project work by providing all the necessary theoretical details and his own experimental expertise for developing a good system. I would also like to thank Dr G Santosh Babu and Mr Ram Prasad for their technical assistance in setting up the experiment and making sure that the instruments are in top working condition. I would like to extend my heartfelt esteems to all staff in laboratory for their timely support. Finally, I would like to express my sincere gratitude to my lab partner Mr Divyanshu Gupta for not only staying by my side and offering his experimental expertise in optical alignment but also providing the necessary emotional and mental support during frustrating moments (like perfect alignment of the Nd:YAG laser which took more than a week for completion) of the experiment.

12 Appendix-Observation Tables

12.1 Absorption Spectrum of Pump Diode LASER:

V_{max}^{sat}= 9.7 Volts

		Set 1		Set 2	
S. No.	**Diode Laser Temperature ^{0}C**	**Observed Voltage (V)**	**Absorption**	**Observed Voltage (V)**	**Absorption**
1	15	3.62000	0.6268041237	3.91000	0.5969072165
2	16	2.89000	0.7020618557	3.03000	0.687628866
3	17	2.56000	0.7360824742	2.54000	0.7381443299
4	18	1.98000	0.7958762887	1.93000	0.8010309278
5	19	1.65000	0.8298969072	1.45000	0.8505154639
6	20	1.23000	0.8731958763	1.18000	0.8783505155
7	21	1.04000	0.8927835052	1.09000	0.887628866
8	22	0.88200	0.9090721649	1.10000	0.8865979381
9	23	1.19600	0.8767010309	1.38000	0.8577319588
10	24	1.89800	0.8043298969	2.0600	0.787628866
11	25	3.30000	0.6597938144	3.30000	0.6597938144
12	26	4.92000	0.4927835052	4.39000	0.5474226804
13	27	6.08000	0.3731958763	6.10000	0.3711340206
14	28	6.87000	0.2917525773	7.27000	0.2505154639
15	29	8.10000	0.1649484536	8.24000	0.1505154639
16	30	8.36000	0.1381443299	8.66000	0.1072164948
17	31	8.12000	0.1628865979	8.40000	0.1340206186
18	32	7.64000	0.212371134	7.59000	0.2175257732
19	33	6.95000	0.2835051546	7.06000	0.2721649485
20	34	6.22000	0.3587628866	6.33000	0.3474226804
21	35	4.89000	0.4958762887	4.86000	0.4989690722

12.2 Wavelength and temperature dependence of Diode Laser

S. No.	Injection current (mA)	Temperature of Diode Laser (^{0}C)
1	510	15
2	470	16
3	430	17
4	390	18
5	350	19
6	315	20
7	270	21
8	240	22
9	210	23
10	160	24
11	120	25
12	80	26
13	40	27

12.3 Laser Diode Output Power

Distance of optical power meter from the Nd: YAG crystal is D= 44 cms.

S. No.	Injection current (mA)	Power of Diode Laser(mW)
1	140	0.2
2	120	0.09
3	110	0.06
4	100	0.04
5	90	0.03
6	80	0.02
7	70	0.02
8	60	0.01
9	50	0.01
10	40	0.01
11	30	0.01
12	20	0
13	10	0
13	0	0

S. No.	Injection current (mA)	Power of Diode Laser(mW)	S. No.	Injection current (mA)	Power of Diode Laser(mW)
1	700	80.6	29	419	40.6
2	688	78.8	30	410	39.3
3	679	77.2	31	400	38
4	670	76	32	389	36.2
5	660	74.7	33	380	34.9
6	650	73.4	34	370	33.6
7	638	71.1	35	359	31.8
8	628	70.4	36	350	30.5
9	621	69.5	37	341	29.2
10	609	67.7	38	330	27.4
11	599	66.4	39	321	26.1
12	590	65	40	309	24.4
13	583	63.7	41	301	23.1
14	570	62.4	42	290	21.4
15	561	61	43	279	19.7
16	550	59.7	44	270	18.36
17	541	58.4	45	260	16.8
18	529	56.6	46	250	15.15
19	519	55.2	47	241	13.92
20	509	53.9	48	230	12.34
21	500	52.5	49	221	10.71
22	491	51.2	50	210	9.18
23	481	49.8	51	200	7.61
24	469	48	52	190	6.03
25	460	46.7	53	181	4.55
26	450	45.4	54	171	3.17
27	441	44	55	160	1.6
28	429	42.2	56	149	0.47

12.4 Measurement of Threshold and Output of Nd: YAG Laser:

Distance of optical power meter from the Nd: YAG crystal is D= 44 cms.

S. No.	Injection current (mA)	Power of Laser(mW)	S. No.	Injection current (mA)	Power of Laser(mW)
1	700	33.72	23	260	8.6035
2	680	33.33	24	240	7.1055
3	660	32.76	25	220	5.5525
4	640	32.585	26	200	4.0955
5	620	32.145	27	180	2.754
6	600	31.425	28	160	1.1795
7	580	30.58	29	150	0.603
8	560	29.71	30	140	0.1138
9	540	28.84	31	130	0.0415
10	520	27.7	32	120	0.03
11	500	26.43	33	110	0.02
12	480	25.32	34	100	0.02
13	460	24.055	35	90	0.01
14	440	22.77	36	80	0.01
15	420	21.03	37	70	0.01
16	400	19.83	38	60	0
17	380	18.275	39	50	0
18	360	16.54	40	40	0
19	340	15.06	41	30	0
20	320	13.53	42	20	0
21	300	11.89	43	10	0
22	280	10.125	44 3	0	0

12.5 Nd:YAG Power output at various pump Wavelengths:

Distance of optical power meter from the Nd: YAG crystal is D= 29 cms.

S. No.	Diode Laser Temperature (^{0}C)	Power (mW) at 600 mA	Power (mW) at 470 mA	Power (mW) at 320 mA
1	15	27.67	15.28	8.944
2	16	31.84	15.71	8.417
3	17	35.29	15.54	8.210
4	18	39.33	15.40	8.773
5	19	42.59	15.32	9.285
6	20	43.54	18.81	7.478
7	21	44.41	22.61	8.814
8	22	45.26	27.02	11.13
9	23	44.75	32.69	15.65
10	24	43.34	36.51	18.72
11	25	41.66	37.10	19.50
12	26	40.14	37.09	20.00
13	27	41.12	36.41	20.10
14	28	44.14	35.83	20.22
15	29	45.63	35.5	20.00
16	30	49.84	34.23	19.67
17	31	52.12	33.49	19.55
18	32	54.12	32.28	19.03
19	33	55.87	30.78	18.48
20	34	56.59	29.67	17.85
21	35	57.39	29.17	17.34

12.6 Second Harmonic Generation using KTP crystal :

S. No.	Power of Fundamental wave(mW)	Power of SHG (mW)
1	28.43	92.268
2	27.63	87.049
3	26.46	78.402
4	24.5	67.902
5	23.42	68.306
6	21.80	57.886
7	20.01	48.358
8	18.16	36.724
9	16.30	31.896
10	14.80	25.177
11	14.03	24.301
12	12.94	19.792
13	12.00	17.604
14	11.07	14.537
15	10.13	12.245
16	9.332	10.765
17	8.549	8.420
18	7.642	6.783
19	6.961	5.582
20	6.196	4.543
21	5.390	3.596
22	4.650	2.427
23	3.918	1.909
24	3.100	1.181
25	2.509	0.702
26	1.828	0.396
27	1.180	0.170
28	0.9741	0.111
29	0.589	0.042
30	0.300	0.011
31	0.100	0.001

12.7 Spiking:

At Modulation = 50

S. No.	Injection Current(mA)	Time Delay(μs)
1	150	640
2	160	420
3	170	332
4	180	296
5	190	272
6	200	256
7	210	236
8	220	228
9	230	220
10	240	212
11	250	200
12	260	194
13	270	188
14	280	182
15	290	178
16	300	174
17	310	170
18	320	166
19	330	164
20	340	160
21	350	158
22	360	154
23	370	152
24	400	146
25	430	140
26	460	136
27	490	132
28	520	126
29	550	124
30	580	120
31	610	118
32	640	114
33	670	111
34	700	108

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